

PONTIFICIA UNIVERSIDAD CATOLICA DE CHILE
SCHOOL OF ENGINEERING

DESIGN OPTIMIZATION OF A REINFORCED CONCRETE FRAME BUILDING TO MINIMIZE THE DRIFT

PREVIEW

ORLANDO DANIEL ARROYO AMELL

Thesis submitted to the Office of Research and Graduate Studies in partial fulfillment of the requirements for the Degree of Master of Science in Engineering (or Doctor in Engineering Sciences)

Advisor:
SERGIO GUTIÉRREZ CID

Santiago de Chile, november, 2016

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A Dios, mis padres y amigos, por su apoyo en el camino.

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RESUMEN

Los marcos de hormigón armado son un sistema estructural utilizado en todo el mundo, especialmente en los países en desarrollo, debido al costo relativamente bajo del sistema en comparación con otras alternativas. Como reflejo de esta ubicuidad, los investigadores han propuesto métodos de optimización basados en diferentes marcos de optimización, que son capaces de proporcionar buenas soluciones para el problema de optimización, sin embargo, su uso en la práctica por parte de los ingenieros estructurales ha sido escaso. Esta situación se ha producido debido a que estos métodos utilizan modelos estructurales no lineales para evaluar el desempeño sísmico y que son difíciles de implementar para el ingeniero estructural promedio. Además, éstos no pueden proporcionar resultados dentro de los plazos utilizados en la práctica de la ingeniería. Esta investigación presenta un método de optimización para mejorar el rendimiento sísmico de edificios de marcos de hormigón que utiliza la optimización de frecuencias propias para formular el problema y optimización geométrica y el método de homogenización de proponer algoritmos de solución eficientes. El método se aplicó para optimizar edificios con varias configuraciones de altura y planta. Los beneficios de optimización se evaluaron mediante análisis tiempo historia no lineales los análisis aplicados a modelos de fibras de los edificios, cuyos resultados se utilizan para llevar a cabo evaluaciones basadas en la ingeniería sísmica basada en desempeño. Los resultados muestran que la aplicación del método reduce las derivas y desplazamientos de entrepiso en los pisos inferiores y que la fragilidad a colapso de los edificios se reduce. Además, los edificios optimizados tienen una distribución más uniforme de la deriva a lo largo de la altura de los edificios, como consecuencia de la distribución de resistencia que resulta de la aplicación del método. El trabajo futuro en este tema puede llevarse a cabo investigando la aplicación de la optimización de frecuencias propias para marcos de acero, así como evaluar si la formulación del problema se puede extender para incluir más variables de diseño sin comprometer la simplicidad en la formulación y el rendimiento computacional.

Palabras Claves: Marcos de hormigón armado, Optimización de frecuencias propias, ingeniería sísmica basada en desempeño, homogenización, optimización geométrica.

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ABSTRACT

Reinforced concrete frames (RCF) are a widely used structural system around the world, especially in developing countries, due to the system's relatively low cost compared to other alternatives. Reflecting this ubiquity, researchers have proposed optimization methods for RCF based on different optimization frameworks, which are capable of providing good solutions to the optimization problem, however, their usage by practicing structural engineers has been scarce. This situation has occurred because these methods use nonlinear structural models to evaluate the seismic performance and they are difficult to implement for the average structural engineer. Besides, they cannot provide results within the time frames used in engineering practice. This research presents an optimization method to improve the seismic performance of RCF buildings that uses eigenfrequency optimization to formulate the problem and geometric optimization and the homogenization method to propose efficient solution algorithms. The method is applied to optimize buildings with several height and plan configurations. The optimization benefits are assessed using nonlinear time history analyses applied to fiber models of the buildings, whose results are used to conduct evaluations based on performance based earthquake engineering. The results show that the method application reduces the story drifts and displacements in the bottom stories and that the buildings collapse fragility is reduced. Furthermore, optimized buildings have a more uniform distribution of the drift along the buildings' height as a consequence of the strength distribution that results from the application of the method. Future work in this topic can be pursued investigating the application of eigenfrequency optimization for steel frames and evaluate if the problem formulation can be extended to involve more design variables without compromising the simplicity in the formulation and the computational performance.

Keywords: Reinforced concrete frames, Eigenfrequency optimization, Performance based earthquake engineering (PBEE), homogenization, geometric optimization

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EXECUTIVE RESUME

This thesis proposes a method to minimize the drift of reinforced concrete frames (RCF) based on eigenfrequency optimization. This work has two major scientific contributions: first, it demonstrated that eigenfrequency optimization can be used as a framework to formulate the seismic optimization problem for reinforced concrete frames. Second, it proposed an efficient algorithm to solve the optimization problem based on geometric optimization and on the homogenization method.

This document is structured in four chapters. The first one corresponds to an introductory chapter, describing the problem, its objectives, hypothesis, methodology and results.

Chapters 2 to 4 correspond to three ISI papers that present the results of this work to the scientific community. The first paper was published in the Journal of Computing in Civil Engineering and it describes the eigenfrequency optimization as a framework, proposing a simplified solution method based on geometric optimization. The second paper is accepted for publication in Engineering Optimization and it presents the eigenfrequency optimization based on homogenization for optimizing the seismic design of RCF. The third paper presents an in-depth investigation based on performance based earthquake engineering, of the seismic improvements obtained by using the method proposed in the second paper. It is the result of the collaboration with Dr. Abbie Liel during the author's research internship at the University of Colorado Boulder. At the time of presenting this document, this paper is under review. A fourth, practical paper is being elaborated based on design recommendations extracted from the third paper, which will be submitted to a suitable journal.

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1. INTRODUCTION

1.1 Context

Reinforced concrete frames (RCF) are a widely used structural system around the world, especially in developing countries, due to the system's relatively low cost compared to other alternatives. Reports from the World Housing Encyclopedia indicate that RCF account for about 75% of the building stock in Turkey (WHE Report 64), 60% in Colombia (WHE Report 11) and 80% in Mexico (WHE Report 115).

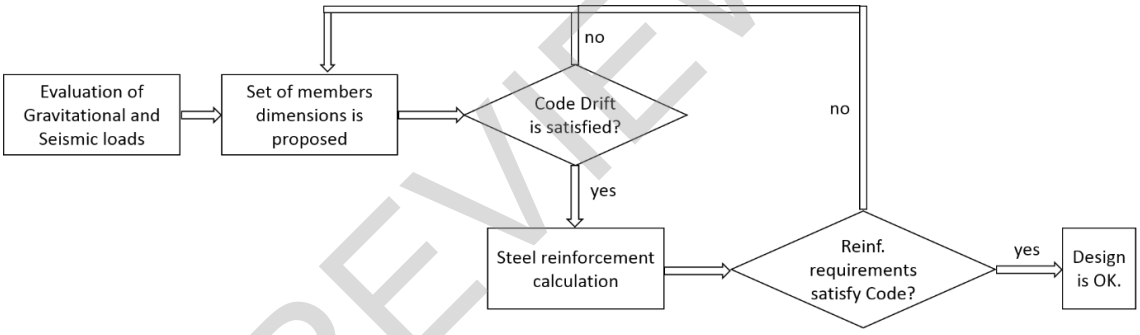


Figure 1-1. The typical design process for a reinforced concrete moment resisting frame uses an iterative process to calculate the column and beam (members) dimensions and steel reinforcement that satisfy the code provisions.

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Traditionally, these buildings are designed by structural engineers following an iterative process that involves three steps (Figure 1-1). First, gravitational and seismic loads are evaluated following the procedures stated by design codes. Then, they calculate a set of columns and beams dimensions that satisfy the elastic drift limit imposed by the code. Finally, steel reinforcement is calculated to satisfy the codes' strength and ductility requirements. In few cases, when elements dimensions are small and they exceed the maximum allowable steel, it is necessary to adjust the columns and beams dimensions to meet these requirements. A design is said to satisfy the code when it fulfills the drift limit

and the strength and ductility requirements. Due to the rapid pace of structural design offices, the customary practice is that a design is considered as final when it satisfies the code and conforms to what previous experience has taught. Because of this practice, most RCF buildings have suboptimal seismic performance.

To address this issue, researchers have proposed optimization methods (e.g. Zou, et al. 2007; Li & Liu, 2010; Li et al. 2010; Khatibinia et al. 2013; Bai et al. 2016; Hajirasouliha et al. 2012) for RCF based on different optimization frameworks (Fragiadakis & Lagaros, 2011), such as deterministic based optimization (DBO) and reliability based optimization (RBO), where the cost of the structure is considered as the objective function and the code requirements are considered as constraints. These methods have demonstrated being effective to solve the optimization problem; however, their usage among engineers have been scarce because they do not have enough computational performance and feasibility of implementation in a practical environment.

The fact that engineers demand seismic optimization methods that are easy to implement and computationally efficient have been acknowledged by researchers, who have introduced methods with improved solution techniques (Hoffman & Richards, 2014), simplified methods for seismic evaluation (Zacharenaki et al, 2013) and where they explicitly consider code constraints (Zou et al., 2007). Although these efforts have yielded improvements in the computational cost, solving RCF seismic optimization problems still takes several hours or even days in consumer-level computers (Zacharenaki et al., 2013). More importantly, these methods still demand the development of complex nonlinear structural models and therefore, they have not captured the attention of engineers. A more detailed discussion about the need to develop seismic optimization methods with an

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orientation to being useful in practice is presented at the end of this chapter, corresponding to an opinion paper written during the development of this thesis.

1.2 Objective and Hypothesis

This thesis is developed with the objective to propose a seismic optimization method that finds the dimensions of columns and beams that minimize the drift in a RCF building, subjected to a constraint in the total volume of the building. The method must have the following characteristics:

1. Optimized buildings must have better seismic performance than traditionally designed buildings, both in the elastic and inelastic regimes.
2. Buildings must be within the scope of existing seismic design codes.
3. High computational performance and feasibility of implementation.
4. The construction complexity of resulting building designs should not increase significantly, nor its analysis and design.

To meet these goals, this work develops an optimization procedure to improve the seismic performance of RCF buildings that uses an eigenfrequency optimization framework for the problem statement and which can be solved using the homogenization method. The hypothesis of this thesis is that eigenfrequency optimization can provide a solid framework to optimize the seismic design of RCF buildings.

1.3 Summary of theoretical principles

This section presents a short description of the theoretical background in this thesis. A more detailed discussion is given in Arroyo and Gutiérrez (2017) and in the third chapter of this document.

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1.3.1 Eigenfrequency Optimization

Topology optimization describes a class of optimization that addresses the problem of determining the optimum distribution of material within a domain. The two most common objectives for topology optimization are the minimum compliance and the eigenfrequency optimization. In eigenfrequency optimization, one or more structural frequencies are optimized. Eigenfrequency optimization has seen several applications in structural engineering, such as stiffness maximization of beam-column connections (Lee et al. 2012) vibration reduction in truss structures (Senba et al. 2013) and the maximization of the fundamental eigenfrequency of geometrically nonlinear beams (Yoon, 2010).

Several solution methods have been proposed for topology optimization problems, including the solid isotropic material with penalization (SIMP) method (Bendsoe and Sigmund, 2003) evolutionary algorithms, and the full homogenization method (Bendsoe and Kikuchi, 1988). The nature of the topology optimization problem can lend itself to solution methods that start with a single initial solution that is gradually improved (*e.g.* SIMP and full homogenization), offering outstanding computational performance that can be programmed to solve 3D problems even on devices with modest computational power such as smartphones and tablets (Nobel-Jørgensen et al. 2014).

The method proposed in this work uses eigenfrequency optimization for the problem formulation and the homogenization method as solution technique. The application of eigenfrequency optimization for RCF makes sense, because during an earthquake the response of a building is heavily dependent on the smallest natural frequencies of the structure, thus, it is reasonable to optimize these structural properties. This dominating

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nature of the smallest natural frequencies in the structural response has been used to develop efficient analyses procedures, such as the modal pushover analysis (Chopra et al., 2004; A. K. Chopra & Goel, 2002); moreover, several standards (ASCE 7-10; ASCE 41-06) acknowledge that the seismic behavior in structures without important torsional irregularities, is controlled by the first natural frequency, allowing the use of the equivalent lateral force procedure to evaluate the seismic demands on structural members.

In addition to optimizing a property that has a direct influence on the seismic performance of the structure, eigenfrequency optimization offers several advantages. First, the objective function is calculated based on the elastic model of the building, resulting in low computational costs. Second, it provides easiness and flexibility in terms of the problem formulation, as it allows for a discrete statement in terms of classical structural analysis (SA), and it can also be stated in a continuous formulation based on partial differential equations (PDEs), which is used here to solve the problem with the homogenization method.

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1.3.2 Problem formulation based on partial differential equations

Eigenfrequency optimization is formulated using PDEs as follows:

$$
\begin{array}{lrcll}
\text{Max} & \omega_n(\boldsymbol{\rho}) & & & \\
\text{subject to} & \boldsymbol{\rho} & \geq & \boldsymbol{\rho}_{\min} & \\
& V(\boldsymbol{\rho}) & = & V_0 & \\
& -\text{div}\left(Ce(u)\right) & = & \omega_n^2 \boldsymbol{\rho} u & \text{in } \Omega \\
& Ce(u)n & = & 0 & \text{on } \Gamma_N \\
& u & = & 0 & \text{on } \Gamma_D
\end{array}
$$

Here, $\boldsymbol{\rho}$ represents density of material, which should be greater than a function $\boldsymbol{\rho}_{\min}$ in the domain Ω, and $V(\boldsymbol{\rho}) = V_0$ means that the total amount of material must be equal to a predefined volume V_0. The last three constraints mean that ω_n is an eigenfrequency of the structure.

As a result of using the material density $\boldsymbol{\rho}$ as the optimization variable, this formulation requires $\boldsymbol{\rho}_{\min}$ and Ω to be defined in terms of the lower and upper bounds of the dimensions of the structural members. Once the optimal density of material $\boldsymbol{\rho}_{opt}$ has been obtained, it needs to be expressed as column and beam dimensions. The steel reinforcement to be used is calculated after the optimization.

The main advantage of the PDE formulation is that it lends itself to solution methods that work with a single initial solution that is gradually improved, like the SIMP (Bendsoe & Sigmund, 2003) and the full homogenization (Tartar, 2009) methods, offering higher computational efficiency and the associated possibility of having a larger search space.

Details about the solution method used in this work, as well as the proposed computational algorithm are presented in Chapter 3.

1.4 Results and scientific contribution

1.4.1 Eigenfrequency optimization as a framework for RCF seismic optimization.

The first scientific contribution of this work is the demonstration that eigenfrequency optimization can be used as a framework to formulate the seismic optimization problem for reinforced concrete frames. This is the result of the first stage of the research and it is

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covered in the second chapter of this document, which is published in the ASCE Journal of Computing in Civil Engineering.

In this publication, the eigenfrequency optimization is proposed for optimizing the column dimensions of RCF buildings. A surrogate model for the buildings based on a membrane is considered (Figure 1-2). The surrogate model to use consists in viewing the building in elevation as a membrane of variable thickness, then optimizing on the thickness to maximize the first eigenfrequency of the membrane and then translating this thickness into depths of the columns of the frames to be used in the building. If the building is either not regular or not slender, the proposed surrogate model is not appropriate for eigenfrequency optimization.

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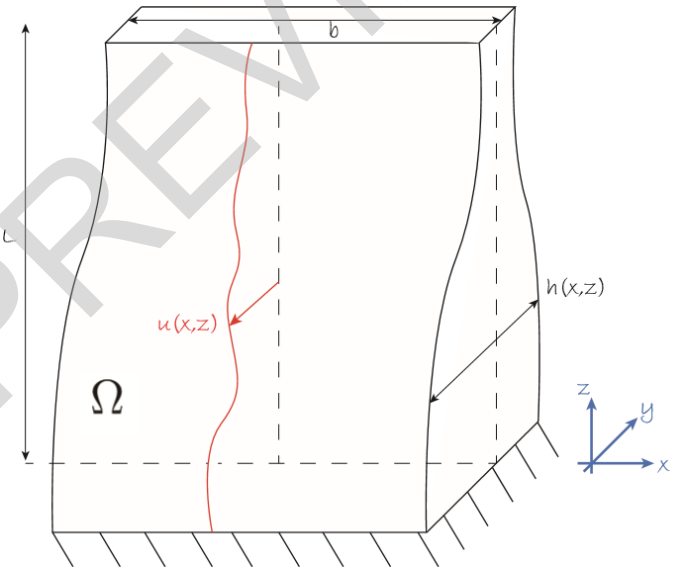


Figure 1-2. Two dimensional model of a membrane

The simplicity of the surrogate model allowed to use geometric optimization to solve the problem (more details in Chapter 2), which was applied to a 10 story building (Figure 1-3).

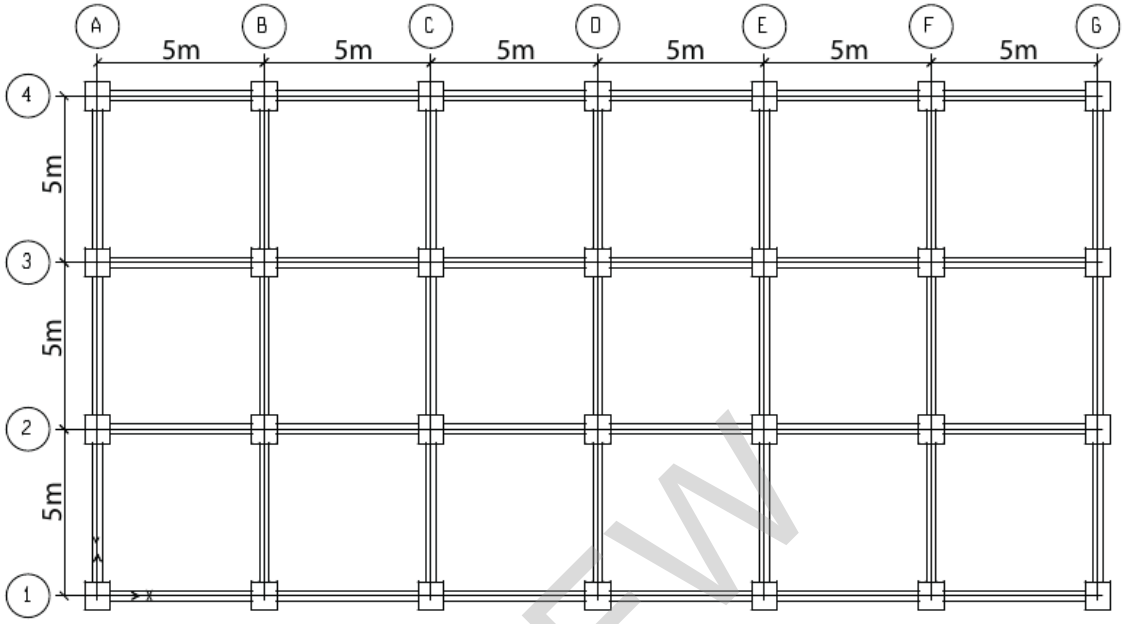


Figure 1-3. Ten story regular building used for demonstrating the applicability of eigenfrequency optimization

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The optimization of dimensions gave the results shown in the blue line of figure 1-4, which were rounded to the closest multiple of 5cm in order to accommodate with construction practices. The total time invested for the optimization is 27s running on a desktop computer with an Intel Core i5 3570K (3.4GHz) with 8GB of RAM.

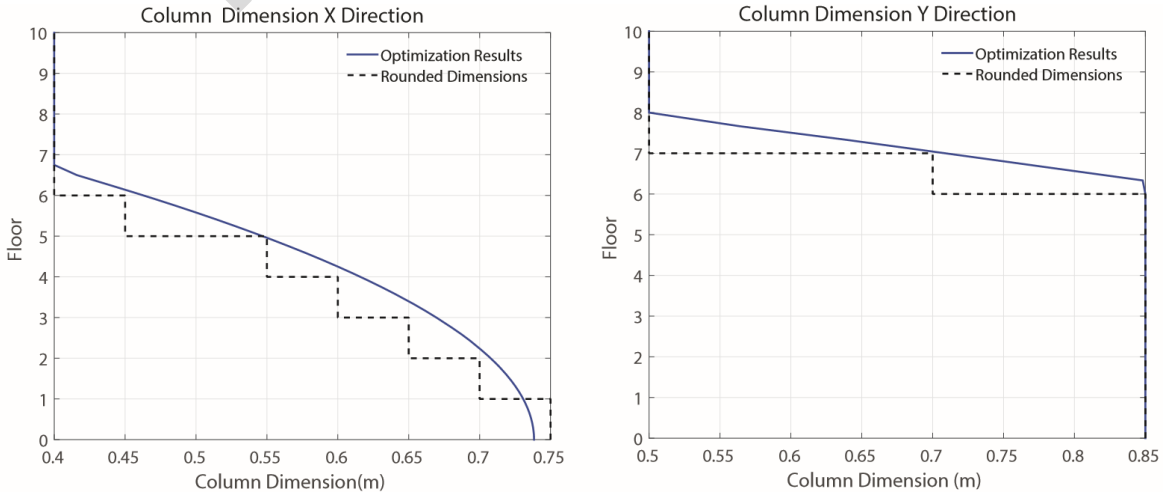


Figure 1-4. Optimized and rounded dimensions for the 10 story building.

The seismic performance of the optimized building is compared using a pushover analysis with a traditional building designed following the procedure described in figure 1-1. The details of the nonlinear models are presented in Chapter 2, and the results of this comparison is shown in figure 1-5, where it can be observed that the optimized building has an overstrength that is 22.3 and 22.4% higher than the baseline structure for the x- and y-directions, respectively. Similarly, the ductility factor is 31.2% greater in the x-direction and 10.2% in the y-direction.

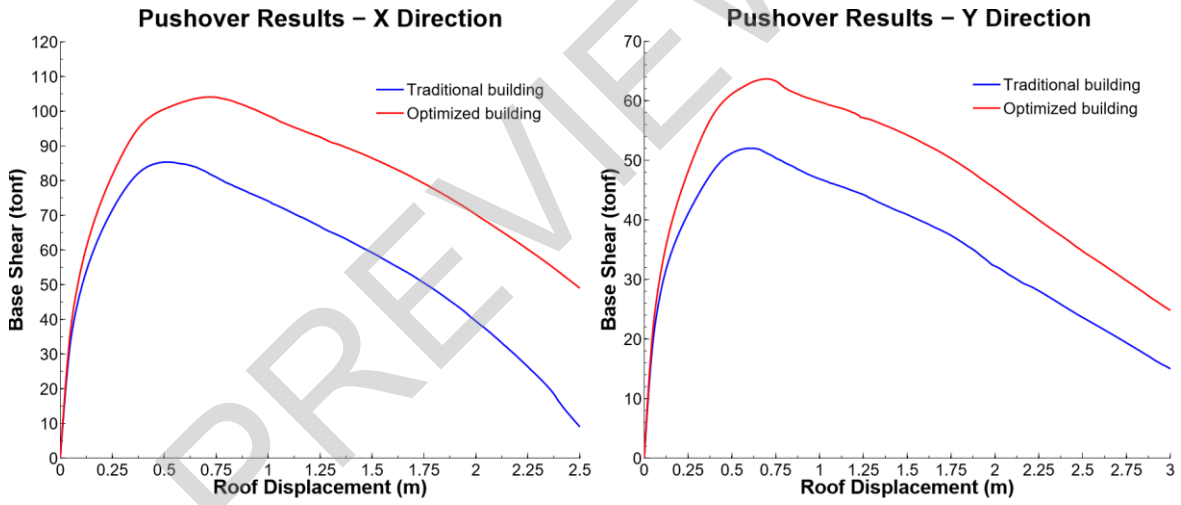


Figure 1-5. Pushover results for the 10 story building.

Further details about these results can be found in Chapter 2 of this document, but to sum them up, it can be stated that the optimized building shows an improved behavior with respect to the baseline in three key areas: elastic drift, overstrength, and ductility. What is more, all these improvements come with no additional expenses associated with material costs, because the volume of the optimized building is almost the same as that of the traditional one.

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1.4.2 Eigenfrequency optimization with full homogenization

The second contribution of this work is proposing an optimization method that uses eigenfrequency optimization with the full homogenization method to determine the optimum member dimensions of reinforced concrete frames.

In this stage, eigenfrequency optimization is formulated as described in section 1.3.2, which requires solving the problem using the finite element method. In this formulation, an optimization domain Ω must be defined based on the maximum desired dimensions for beams and columns. In this domain, the columns and beams dimensions are expressed as a material density ρ, therefore, the minimum dimensions are expressed as $\rho_{\min}$. As previously discussed, this formulation allows to use efficient optimization techniques.

In this work, the full homogenization (FH) method is used as a solution technique for this approach. A description of the mathematical basis of the FH method for a two dimensional case and a computational algorithm to solve the problem are presented in Chapter 3. A more complete description of the method can be found in (Allaire, 2002) and, more specifically, for eigenfrequency optimization in (Allaire et al., 2001). Being this a first study on eigenfrequency optimization for RCF based on full homogenization, this work focuses only on the first eigenfrequency.

The proposed method is used to determine the column and beam dimensions that optimize the seismic performance of the building in figure 1.3. The building was first designed following the traditional procedure according to the ASCE 7 and the ACI 318 for the state of California (37.38ºN, 121.88ºW) using ETABS v13.1.2 (Habibullah, 1997). After that, it was optimized using the method proposed in this thesis, with the results shown in the following table:

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Table 1-1. Dimensions for 10-story buildings (all dimensions are given in cm)

	Optimized building				Traditional building	
	Columns		Beams		Columns	Beams
Story	Inner	Outer	X Dir	Y Dir	Both	Both
10	50x60	50x60	35x35	35x35	55x75	35x40
9	55x65	55x65	35x35	35x35	55x75	35x40
8	55x70	55x70	35x40	35x40	55x75	35x40
7	60x70	55x70	35x40	35x40	55x75	35x40
6	60x75	55x75	35x40	35x40	55x75	35x40
5	60x75	55x75	35x40	35x40	55x75	35x40
4	65x80	60x80	35x45	35x45	55x75	35x40
3	70x80	65x80	35x45	35x50	55x75	35x40
2	70x80	70x80	35x45	35x50	55x75	35x40
1	70x85	70x80	35x45	35x45	55x75	35x40

The seismic performance of the buildings is compared using OpenSees by means of a pushover analysis and an incremental dynamic analysis of the record set of the FEMA P-695.

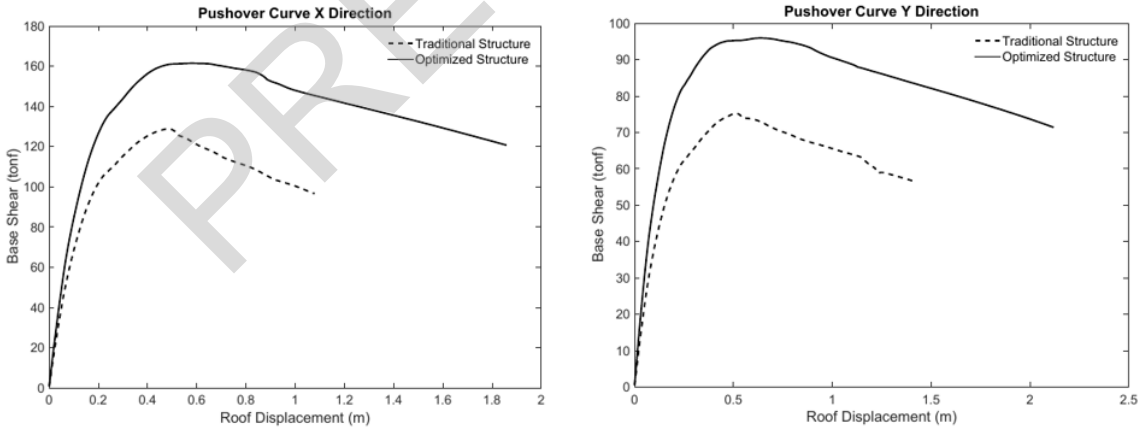


Figure 1-6. Pushover results for the 10 story building.

The pushover results (figure 1-6) show that the optimized building has an overstrength that is 23.8% and 27.9% higher than the baseline building for the X and Y direction, respectively. On the other hand, the ductility factor sees a notable improvement, as it is 81.9% greater in the X direction and 98.2% in the Y direction.

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The performance improvement shown in the Pushover is verified using Nonlinear Time History Analysis. We start by looking at the displacement response for a scale factor of 1.0 in figure 6, where it can be appreciated that the median displacement of the optimized building is smaller than the baseline, in a range that varies from 45% for the first floor in the X direction, down to 10% for the tenth floor in the Y direction.

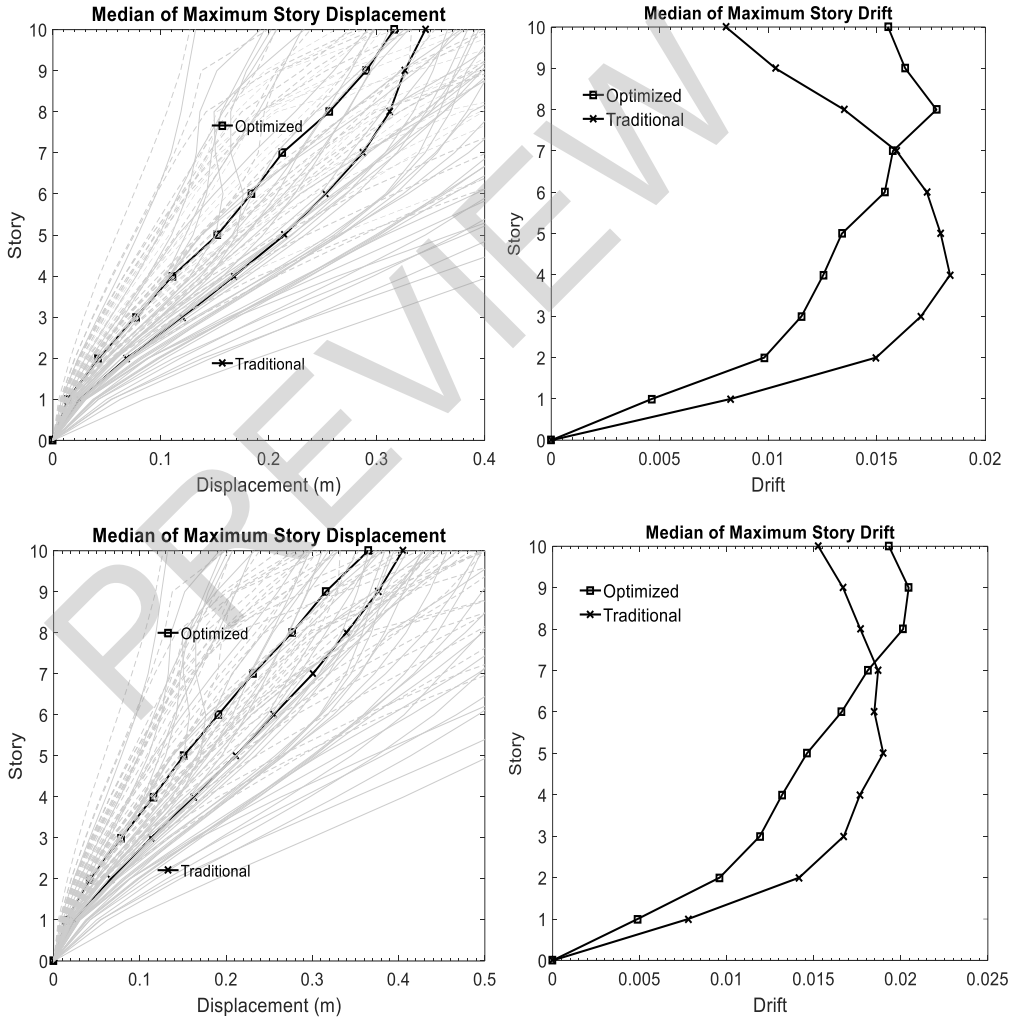


Figure 1-7. The displacement and drift responses for a scale factor of 1.0 are smaller for the optimized building, especially in the bottom stories. Top: X direction. Bottom: Y direction. Dotted and continuous gray lines represent individual ground motion results for the optimized and traditional buildings, respectively.

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